Cairo University	Academic Year:	2021/2022	Term Exam:	Second Term	
	Department:	Electronics & Electrical Communications	Course Code:	ELC458	
	Date:	June 2022	Course Title:	Selected topics in Digital Integrated Circuits	
	Time:	2 Hours	Full Mark:	70	

1- If you could use a clock with jitter only or skew only, which would you choose and why?

I would choose clock skew because it is deterministic not stochastic. So, once I knew the skew value, I can place different buffers in the clock path to ensure that it reaches all flip flops at the same time.

2-Write an expression for a clock period suffering from jitter and for the hold-time suffering from negative skew

T_clk > Tcq + T_CLB + Tsu + 2*Tj (where Tj is the maximum jitter that can happen)

Th < Tcq + T_CLB + T_skew

3- Write an expression for hold-time under negative jitter and hold-time under positive jitter

Th < Tcq + T_CLB

Hold time is not affected by jitter

4-Write an advantage and a disadvantage for the tree method for clock distribution

Advantage: has less area and accordingly power compared to the grid method. Another advantage is that it minimizes the relative skew between communicating flip flops.

Disadvantage: design dependent

5-Write an expression for the delay of an N-bit ripple carry adder (define all variables you use if they differ from those in lecture)

Tpd = (N-1) * Tcarry + Tsum

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6-Write an expression for the delay of an N-bit carry bypass adder (define all variables you use)

K * Tcarry + (N/K - 1) * Tmux + (K-1) * Tcarry + Tsum

K: number of bits per stage
N: total number of bits
N/K: number of stages

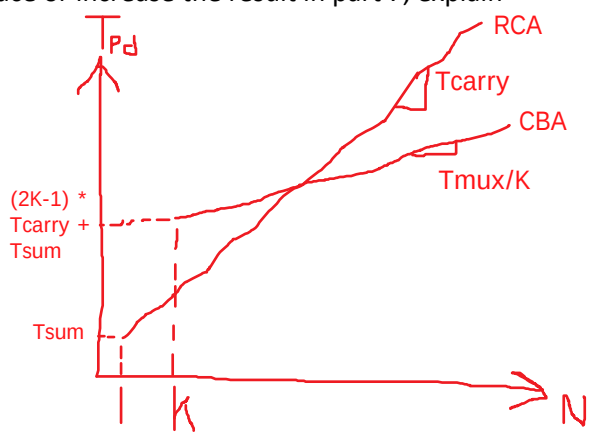
7-At what total number of bits N does a carry bypass adder become more efficient than a ripple carry adder

K * Tcarry + (N/K - 1) * Tmux + (K-1) * Tcarry + Tsum < (N-1) * Tcarry + Tsum

N > K * ((2KTcarry - Tmux) / (KTcarry - Tmux))

8-Does raising the number of bits per block in the carry bypass adder reduce or increase the result in part 7, explain why this happens briefly

Increasing the number of bits per block in the carry bypass adder increases the result in part 7, which means the carry bypass adder becomes more efficient than a ripple carry adder for a larger number of bits. This can be justified using the graph on the right. It may seem that the slope of the CBA is inversely proportional with K so we may expect that increasing K makes the CBA line meets the RCA line at an earlier point and hence N decreases, but this is not true because we must pay attention that the first point on the CBA line is function of K in both x-axis and y-axis. So, increasing K shifts the first point on the CBA line up and to the right with an effect greater than that of the slope being lowered, so the CBA line meets the RCA line at a farther point, which means larger N.



9-Explain why this statement is **not** correct "An asynchronous FIFO will not work because the full and empty flags pass through synchronizers which have a one cycle uncertainty"

It is true that there is a one cycle uncertainty in synchronizers, but this won't cause the full and empty flags to fail. Why? because this one cycle uncertainty affects the full and empty flag in a conservative way. That is, it causes the full flag to be set for an extra cycle while it was supposed to be reset, and causes the empty flag to be set for an extra cycle while it was supposed to be reset. This causes the transmitter to stop sending thinking that the FIFO is full for an extra cycle while it is not, and it causes the receiver not to read thinking that the FIFO is empty while it is not. Both effects are in a conservative direction which will not cause any failure in the operation of the FIFO.

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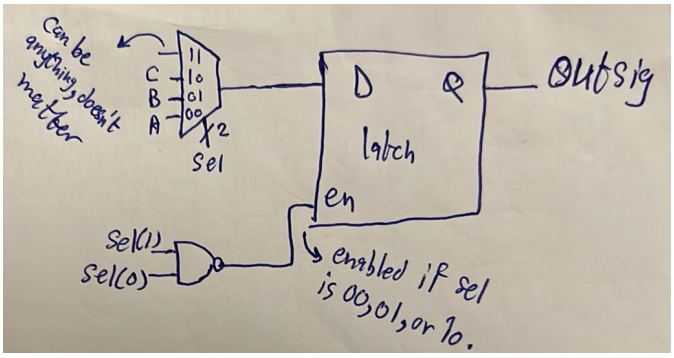
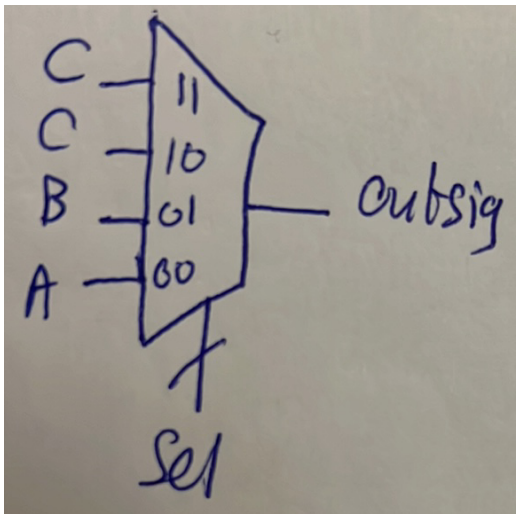
10- Draw an equivalent circuit for these two snippets of code

Circuit 1	Circuit 2

Process(sel, A, B, C)
Begin
  If (sel="00") then
  Outsig <= A;
  Elsif (sel="01") then
  Outsig <= B;
  Elsif (sel="10") then
  Outsig <= C;
  End if;
End process;
	Process(sel, A, B, C)
	Begin
	  If (sel="00") then
	  Outsig <= A;
	  Elsif (sel="01") then
	  Outsig <= B;
	  Else
	  Outsig <= C;
	  End if;
	End process;

Circuit 1	Circuit 2
	

11- Explain why a two-port RAM array is declared as a shared variable while a one port RAM is declared as a signal

A two-port RAM array is declared as a shared variable because it is used in two different processes. VHDL doesn't allow a signal to be accessed from different processes. O the other hand, a one-port RAM is declared as a signal because it is accessed from a single process only.

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12-Technology scaling occurs by a ratio S. However, global wire length scaling is constant (i.e. scales by 1). To control wire resistance, we also scale wire height (thickness) by 1. All other dimensions including wire width and separation scale at the same rate as gate dimensions (S). Find an expression for the percentage increase in interwire capacitance between two adjacent parallel wires of the same length.

Cint = epsilon * L * t_w / L_sep

L --scale by --> 1
t_w --scale by --> 1
L_sep --scale by--> S

Therefore: Cint ---> Cint * S

Increase percentage = (Cint*S - Cint)/Cint * 100 % = (S-1) * 100 %

13-If voltage drops by a ratio U when scaling in question 12, find an expression for current scaling in the wire. Find an expression for current density scaling in the wire.

R --> R * S

V --> V/U

I = V/R

Therefore scaling of I is US (i.e. I --> I/US)

A = W*tw
Therefore wire cross-sectional area scaling is S (i.e. A --> A/S)

Current Density = I/A
Therefore current density scaling is U (i.e. J --> J/U)

13-A local wire is allowed to scale in all dimensions (including length) at a rate similar to gate scaling (S). Find an expression for the percentage increase in wire resistance.

R = L / sigma*W*tw

L --> L/S
W --> W/S
tw --> tw/S

Therefore: R --> R * S

Increase percentage = (R*S - R)/R * 100 % = (S-1) * 100 %

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14- Show that the time to exit metastability is:

$$T_{exit} = \tau \ln(V_{DD}/V_d(0))$$

Start from the sketch of the two inverters as amplifiers

15- The following is a next state process for a state machine. It is written badly. Rewrite it and briefly point out the problem.

```
Process(reset, clk)
Begin
        If clk'event and clk='1' then
                If reset='1' then
                        Next_state <= reset;
                Else
                        Next_state <= current_state;
                End if;
        End if;
End process;
```

```
process(reset, clk)

Begin

  if reset = '1' then
    next_state <= reset_state;
  elsif clk'event and clk = '1' then
    next_state <= current_state;
  end if;

end process;
```

Notes:
1. Using an asynchronous reset is better than a synchronous reset to allow an independent escape for all registers.
2. Naming the reset signal as "reset" as well as the reset state in the enumerator of the states as "reset" might be confusing for the synthesis tool. Better name the state as "reset_state".

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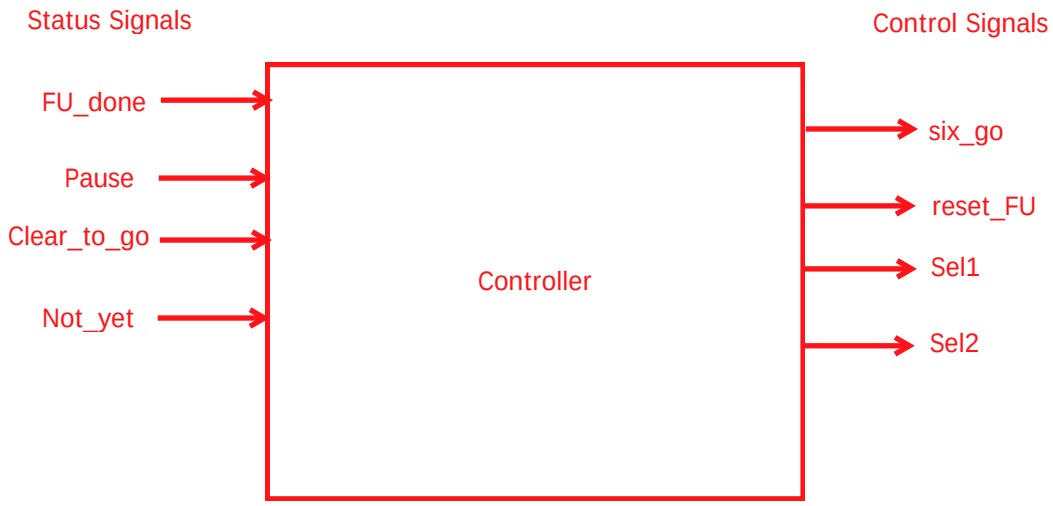
16-Sketch a block diagram showing control signals and status signals of the state machine described by these two processes. Neither process is anywhere near complete, but the code given is enough to give you the information.

Next state process	Output process
Process (FU_done, Pause, Clear_to_go, Not_yet, current_state) Begin ……. End process;	When reset => Six_go <= '1'; reset_FU='1'; Sel1<='001'; Sel2<='002'; ……



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17- Draw the equivalent circuit of the following (nothing is case sensitive):

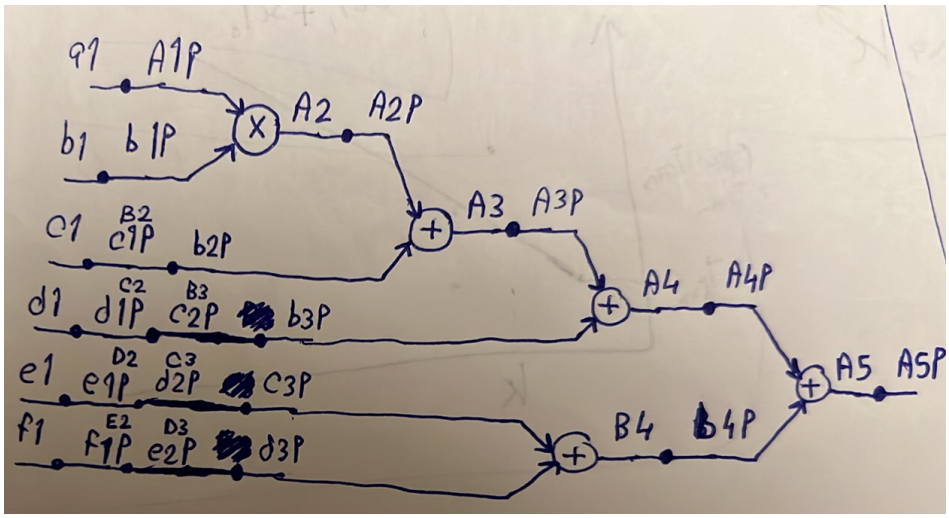
```
Process(clk, reset)
Begin
If reset = '1' then
        A1p<=others=>'0'; b1p<=others=>'0'; c1p<=others=>'0'; d1p<=others=>'0'; e1p<=others=>'0';
        f1p<=others=>'0';
        A2p<=others=>'0'; b2p<=others=>'0'; c2p<=others=>'0'; d2p<=others=>'0'; e2p<=others=>'0';
        A3p<=others=>'0'; b3p<=others=>'0'; c3p<=others=>'0'; d3p<=others=>'0';
        A4p<=others=>'0'; b4p<=others=>'0';
        A5p<=others=>'0';
Elsif clk'event and clk='1' then
        A1p<=a1; b1p<=b1; c1p<=c1; d1p<=d1; e1p<=e1; f1p<=f1;
        A2p<=a2; b2p<=b2; c2p<=c2; d2p<=d2; e2p<=e2;
        A3p<=a3; b3p<=b3; c3p<=c3; d3p<=d3;
        A4p<=a4; b4p<=b4;
        A5p<=a5;
End if;
End process;
```

A2<=a1p*b1p; B2<=c1p; C2<=d1p; D2<e1p; E2<=f1p;

A3<=a2p+b2p; B3<=c2p; C3<=d2p; D3<=e2p;

A4<=a3p+b3p; B4<=c3p+d3p;

A5<=a4p+b4p;



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18- Write an expression for the delay of a stage in a latch loop assuming no slack borrowing

tpd = tdq + t_CLB
tdq = tinv1 + t_T1 + tinv2

19-Write the expression for the condition that needs to be satisfied for total borrowing in every stage of a latch loop so that it is still functional

Total borrowing in every stage < Tclk/2 - tsu

20-For this question assume tcq=tsu=0 for all latches. Also assume that for a FA tsum=5tc. A latch loop consists of three stages. The first is a 4-bit ripple carry adder. The second is an 8-bit ripple carry adder. The clock period T for this loop is 20tc. The third stage is an N-bit ripple carry adder. What is the maximum possible value for N that allows this latch loop to function properly?

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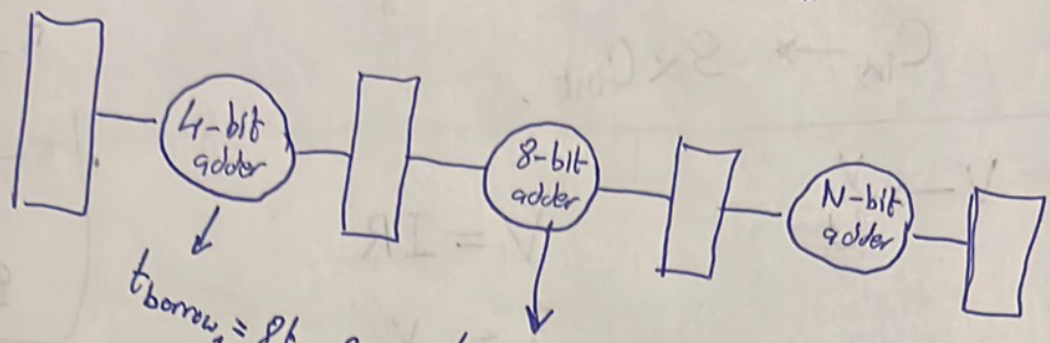
$t_{dq} = t_{su} = 0$

$t_{sum} = 5t_c$

$T = 20t_c$

$t_{pd}|_{\substack{N\text{-bit}\\RCA}} = (N-1)t_c + t_{sum} = (N+4)t_c$

$\therefore t_{pd}|_{\substack{4\text{-bit}\\RCA}} = 8t_c$, $t_{pd}|_{\substack{8\text{-bit}\\RCA}} = 12t_c$



$t_{borrow_1} = 8t_c - \frac{20t_c}{2}$
$= -2t_c$

$\therefore$ Stage 1 doesn't borrow any time from stage 2

$t_{borrow_2} = 12t_c - 10t_c$
$= 2t_c$

$\therefore$ Stage 2 borrows $2t_c$ from stage 3

Condition on stage 3: $\sum t_{borrowed} < \frac{T_{clk}}{2} - \cancel{t_{su}}$ $\rightarrow 0$

$2t_c + t_{borrowed}|_{stage3} < T_{clk}/2$

$2t_c + t_{pd}|_{\substack{N\text{-bit}\\RCA}} - \frac{T_{clk}}{2} < T_{clk}/2 \Longrightarrow 2\cancel{t_c} + (N+4)\cancel{t_c} < 20\cancel{t_c}$ $\therefore \boxed{N_{max} = 14}$

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